Planning EV Infrastructure in Warsaw, Poland

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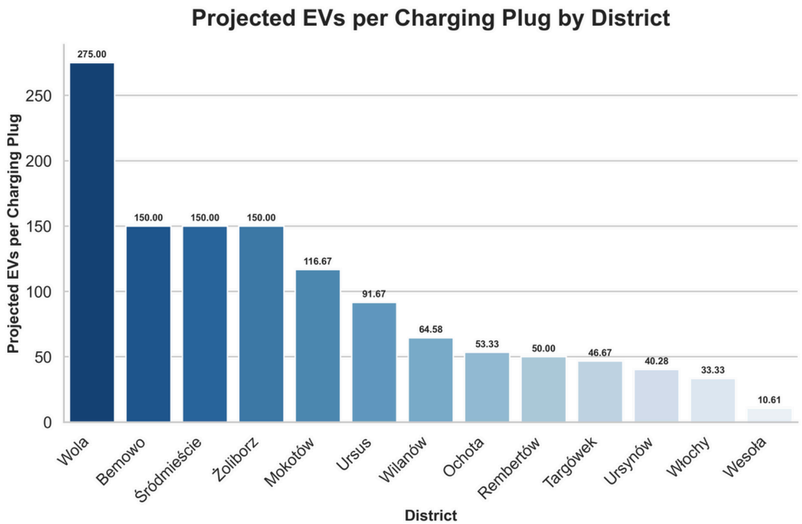
Project Track: Data Science
Majors: Data Analytics & Mathematics

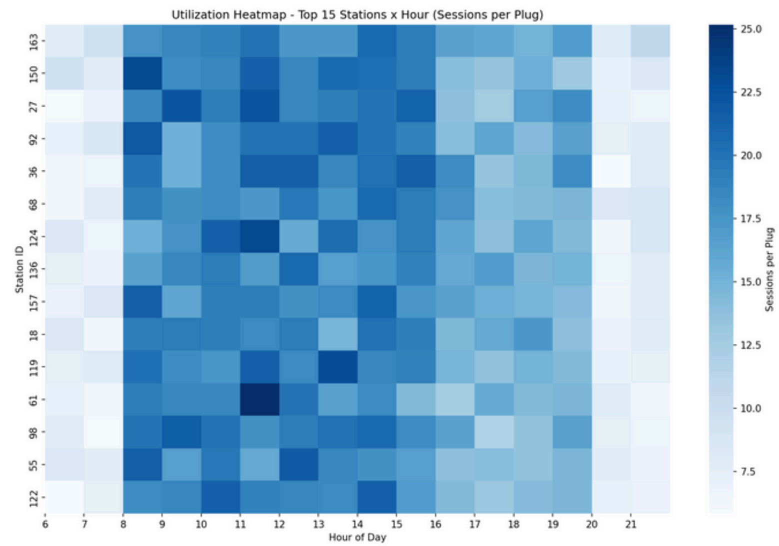
Problem Statement : Public charging station availability is not growing at a fast enough rate (41% growth between 2023 and 2024) to keep up with the growth of electric vehicle (EV) usage (54% growth)[1]. Even where chargers exist, they are not always placed or priced optimally:

Which districts are most at risk of future overload, and where should new plugs be installed to balance current congestion with projected EV growth?

Method:

- Sourced from Kaggle (4 tables: sessions, stations, customers, districts)
- Joined the four datasets into one enriched dataset with hour, day-of-week, sessions-per-plug, EVs-per-plug
- Visualized key usage and capacity patterns to uncover relevant and impactful insights



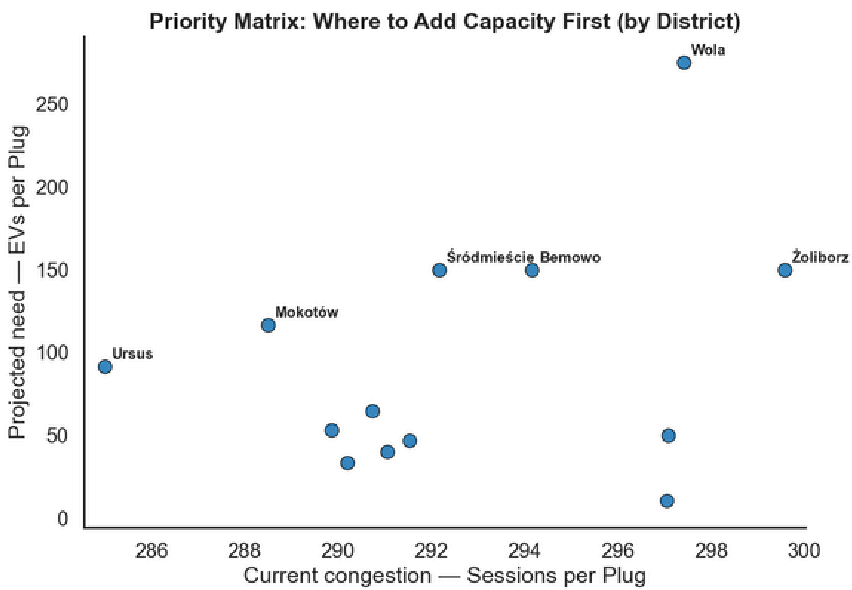


EVs per Plug:

- Capacity is very uneven, with districts like Wola, Bemowo, Śródmieście, etc., having massive plug deficits compared to more efficient districts
- These high-ratio districts are going to be the first places to be overcrowded as EV adoption grows

Utilization Heatmap:

- Each row is one of the 15 most-used stations
- Demand is heaviest from 8 a.m. to 4 p.m., with several stations being consistently dark in that window, having chronic congestion rather than spikes



Results

Main takeaways:

- 4 of the 13 districts have massively disproportionate projected EV to charging plug ratios
- Pricing at each station is the same
- Although low income areas hold about a third of the overall share of stations they hold about 43% of the top 30 most used stations where low income has 13, middle income has 10, and high income has 7

How is this applicable to the real-world?:

- By finding the stations and districts facing congestion, we see that the supply is insufficient and that there is a market for further station development by an existing operator or a new competitor

Why should we care?:

- Using the insights we uncovered, we find that implementing a new station in a district like Wola would not only relieve the area of congestion, but would also guarantee that the station would see a lot of use

Potential Applications:

- Implementing variable pricing based on demand found in the Utilization Heatmap
- Establishing new charging stations, starting in Wola and expanding according to the priority matrix.

LinkedIns, GitHub Repo, & Dataset:

BIG DATA & ANALYTICS ASSOCIATION

1. European Alternative Fuels Observatory (2024)